

SOLUTION

The N-Queens Problem is solved by placing one queen in each row while ensuring that no two queens threaten each other.

1. Represent the chessboard as an $N \times N$ **matrix**.
2. Start placing queens from the first row.
3. For each row, check every column to determine whether it is safe.
4. A position is considered safe if:
 - No queen exists in the same column.
 - No queen exists in the upper-left diagonal.
 - No queen exists in the upper-right diagonal.
5. If a safe position is found, place the queen and move to the next row.
6. If no safe position is available, remove the previously placed queen and try another position (backtracking).
7. Continue this process until all queens are successfully placed.
8. Display the final chessboard configuration representing the solution.